QP Code **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Exam Type EXAMINATION Month YYYY

Semester 5

Discipline Specific Elective (DSE)- MG5DSEPHY300 - SEMICONDUCTOR OPTOELECTRONIC DEVICES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What is cathodoluminescence? [K]/[CO1]
2. Write the k-selection rule for absorption in semiconductors. [K]/[CO1]
3. What is bandtailing? [U]/[CO1]
4. Mention a distinguishing feature in the absorption spectra of a bound exciton in comparison to a free exciton. [U]/[CO1]
5. How does the presence of O and N increase the radiative efficiency of GAP? [A] / [CO2]
6. What is meant by a binary and a ternary LED material? Give one example for each. [K] / [CO2]
7. Why are GaAs/AlGaAs heterostructure lasers used for short-distance communication? [U] / [CO3]
8. What are the factors on which the end losses of heterojunction lasers depend? [U] / [CO3]

9. What are the functions of optically controlled hi-lo switching devices? [U] / [CO4]

10. Explain the need for modulation in communication. [A] / [CO4]

11. Give the schematic illustration of photonic switching characteristics. [A] / [CO4]

12. What is half-wave voltage? [K] / [CO4]

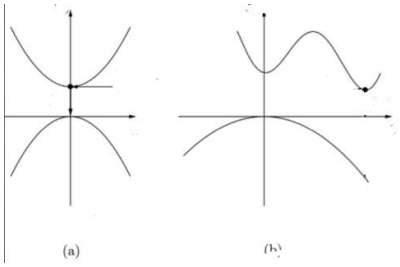
[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

13. 

Identify the direct and indirect semiconductors from the figure. Give reason for your choice. Mark the axes and the bands. [U]/ [CO1]

14. The radiative lifetime is 10 ns and the nonradiative lifetime is 0.5 ns in GaN. Find the total or effective lifetime and the internal quantum efficiency. [U]/CO1]

15. Discuss the various low-energy absorption processes. [K]/[CO1]

16. Compare homojunction and heterojunction LEDs [U] / [CO2]

17. Explain the spectral response and V-I characteristics of LED. [U] / [CO2]

18. With the help of a diagram, explain the stripe geometry of a heterojunction semiconductor laser. [A] / [CO3]

19. Describe the quantum-confined Stark Effect. [U] / [CO4]

20. Draw the block diagram for electro-optic amplitude modulation and explain it. [U] / [CO4]

[6x5 = 30]